



9157 - Probing the Dynamical History of a White Dwarf Planet

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Sydney Jenkins (PI)	Massachusetts Institute of Technology
Dr. Andrew Vanderburg (CoI)	Harvard University
Mary Anne Limbach (CoI)	University of Michigan
Dr. Simon Blouin (CoI) (CSA Member)	University of Victoria
Dr. Peter Gao (CoI)	Carnegie Institution of Washington
Dr. Ryan J MacDonald (CoI) (ESA Member)	University of St Andrews
James Mang (CoI)	University of Texas at Austin
Dr. Caroline Morley (CoI)	University of Texas at Austin
Dr. Kevin Stevenson (CoI)	The Johns Hopkins University Applied Physics Laboratory

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MRS	MIRI Medium Resolution Spectroscopy	(1) WD1856+534
	2	MRS	MIRI Medium Resolution Spectroscopy	(1) WD1856+534
	3	MRS	MIRI Medium Resolution Spectroscopy	(1) WD1856+534
	4	MRS	MIRI Medium Resolution Spectroscopy	(1) WD1856+534
	5	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(1) WD1856+534

ABSTRACT

Most known exoplanets orbit stars that will eventually evolve into white dwarfs. However, little is known about the fate of these post-main-sequence worlds. One of the few known white dwarf planets is WD 1856+534 b, which orbits at just 0.02 au from its host star—close enough that it should

have been engulfed during the star's transition of the main sequence. Two main dynamical pathways have been proposed to explain this: high-eccentricity migration and common envelope evolution. We propose observations to constrain WD 1856 b's dynamical history by searching for signatures of common envelope evolution in the planet's atmosphere. If it did undergo common envelope evolution, it will be the first known planet to have survived engulfment by its star. Regardless of its dynamical history, our program will also probe the atmospheric physics of one of the coldest known planets. These results will provide critical insight into the dynamics of post-main-sequence systems and the atmospheres of cold worlds.

OBSERVING DESCRIPTION

MIRI MRS (31 hours) and NIRSpec (1 hour) observations of WD 1856+534.

Proposal 9157 - Targets - Probing the Dynamical History of a White Dwarf Planet

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WD1856+534	RA: 18 57 39.3437 (284.4139321d)	Proper Motion RA: 240.749 mas/yr	
			Dec: +53 30 33.30 (53.50925d)	Proper Motion Dec: -52.47000008239411 mas/yr	
			Equinox: J2000	Parallax: 0.040393099999999994"	
				Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[White dwarfs]</i> <i>Extended=NO</i>				

Proposal 9157 - Observation 1 - Probing the Dynamical History of a White Dwarf Planet

Observation	Proposal 9157, Observation 1: MRS												Wed Jan 21 00:00:26 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over middle threshold												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	WD1856+534	RA: 18 57 39.3437 (284.4139321d) Dec: +53 30 33.30 (53.50925d) Equinox: J2000				Proper Motion RA: 240.749 mas/yr Proper Motion Dec: -52.47000008239411 mas/yr Parallax: 0.04039309999999994" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[White dwarfs] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F560W	FASTGRPAVG8	10	1	1	222.003	254934				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	100	25	1	Dither 1	4	100	28016.804	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	25	1	Dither 1	4	100	28016.804	

Proposal 9157 - Observation 1 - Probing the Dynamical History of a White Dwarf Planet

Special Requirements	Phase .02 to .3 with period .7039696085 Days and zero-phase 2459038.4358981 HJD
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Proposal 9157 - Observation 2 - Probing the Dynamical History of a White Dwarf Planet

Observation	Proposal 9157, Observation 2: MRS												Wed Jan 21 00:00:26 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over middle threshold												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	WD1856+534	RA: 18 57 39.3437 (284.4139321d) Dec: +53 30 33.30 (53.50925d) Equinox: J2000				Proper Motion RA: 240.749 mas/yr Proper Motion Dec: -52.47000008239411 mas/yr Parallax: 0.04039309999999994" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[White dwarfs] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F560W	FASTGRPAVG8	10	1	1	222.003	254934				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	100	25	1	Dither 1	4	100	28016.804	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	25	1	Dither 1	4	100	28016.804	

Proposal 9157 - Observation 2 - Probing the Dynamical History of a White Dwarf Planet

Special Requirements	Phase .02 to .3 with period .7039696085 Days and zero-phase 2459038.4358981 HJD
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Proposal 9157 - Observation 3 - Probing the Dynamical History of a White Dwarf Planet

Observation	Proposal 9157, Observation 3: MRS												Wed Jan 21 00:00:26 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over middle threshold												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	WD1856+534	RA: 18 57 39.3437 (284.4139321d)				Proper Motion RA: 240.749 mas/yr						
			Dec: +53 30 33.30 (53.50925d)				Proper Motion Dec: -52.47000008239411 mas/yr						
			Equinox: J2000				Parallax: 0.04039309999999994"						
							Epoch of Position: 2000						
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
Category=Star													
Description=[White dwarfs]													
Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F560W	FASTGRPAVG8	10	1	1	222.003	254934				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	100	25	1	Dither 1	4	100	28016.804	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	25	1	Dither 1	4	100	28016.804	

Proposal 9157 - Observation 3 - Probing the Dynamical History of a White Dwarf Planet

Special Requirements	Phase .02 to .3 with period .7039696085 Days and zero-phase 2459038.4358981 HJD On Hold Obs ON HOLD until completion of NIRSpec observation (Obs 5)
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Proposal 9157 - Observation 4 - Probing the Dynamical History of a White Dwarf Planet

Observation	Proposal 9157, Observation 4: MRS												Wed Jan 21 00:00:26 GMT 2026	
	Diagnostic Status: Warning													
Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over middle threshold													
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	WD1856+534	RA: 18 57 39.3437 (284.4139321d)				Proper Motion RA: 240.749 mas/yr							
			Dec: +53 30 33.30 (53.50925d)				Proper Motion Dec: -52.47000008239411 mas/yr							
			Equinox: J2000				Parallax: 0.04039309999999994"							
							Epoch of Position: 2000							
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.														
Category=Star														
Description=[White dwarfs]														
Extended=NO														
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID					
	1	SAME	F560W	FASTGRPAVG8	10	1	1	222.003	254934					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		NO				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	MEDIUM(B)	MRSLONG		FASTR1	100	25	1	Dither 1	4	100	28016.804		
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	25	1	Dither 1	4	100	28016.804		

Proposal 9157 - Observation 4 - Probing the Dynamical History of a White Dwarf Planet

Special Requirements	Phase .02 to .3 with period .7039696085 Days and zero-phase 2459038.4358981 HJD On Hold Obs ON HOLD until completion of NIRSpec obs (Obs 5)
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Proposal 9157 - Observation 5 - Probing the Dynamical History of a White Dwarf Planet

Observation	Proposal 9157, Observation 5: NIRSpec										Wed Jan 21 00:00:26 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	WD1856+534	RA: 18 57 39.3437 (284.4139321d) Dec: +53 30 33.30 (53.50925d) Equinox: J2000			Proper Motion RA: 240.749 mas/yr Proper Motion Dec: -52.47000008239411 mas/yr Parallax: 0.04039309999999994" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[White dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	254934	
Template	HFF Readout Mode				Slit				Subarray			
	false				S1600A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	3							NONE			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	S1600A1	NRSIRS2RAPID	80	1	1	NONE	3	3	3545.1	

Proposal 9157 - Observation 5 - Probing the Dynamical History of a White Dwarf Planet

Special Requirements	Phase .02 to .3 with period .7039696085 Days and zero-phase 2459038.4358981 HJD
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